Supplement-FKG Inequality

- (FKG Inequality I) Let $X_1, \ldots, X_n$ be independent random variables, and let f, g be increasing functions of n variables. Then we have

$$\mathbb{E}[f(X_1,\ldots,X_n)g(X_1,\ldots,X_n)] \geq \mathbb{E}f(X_1,\ldots,X_n)\mathbb{E}g(X_1,\ldots,X_n)$$

<u>Proof</u>

We do it by induction.

For the case $n=1$, monotonicity implies

$$[f(x)-f(y)][g(x)-g(y)] \geq 0$$

Take y as an independent copy of X and take expectation and we have derived the result.

Assume FKG inequality holds for all that are no greater than $n-1$, we now prove it for n.

$$\begin{aligned}\mathbb{E}[f(X_1,\ldots,X_n)g(X_1,\ldots,X_n)] &= \mathbb{E}\left(\mathbb{E}(fg|X_n)\right)\\ &\geq \mathbb{E}\left(\mathbb{E}(f|X_n)\cdot\mathbb{E}(g|X_n)\right)\end{aligned}$$

$\mathbb{E}(f|X_n)$ is an increasing r.v. of X_n

Thus

$$\begin{aligned}\mathbb{E}[f(X_1,\ldots,X_n)g(X_1,\ldots,X_n)] &\geq \mathbb{E}(\mathbb{E}(f|X_n))\cdot\mathbb{E}(\mathbb{E}(g|X_n))\\ &= \mathbb{E}f(X_1,\ldots,X_n)\mathbb{E}g(X_1,\ldots,X_n)\end{aligned}$$

Now we define increasing events.

Assume there is a partial order in Ω

- For $A \in \mathscr{F}$, we call it an increasing event if $\omega \in A \Rightarrow \omega' \in A, \forall \omega' \geq \omega$

By FKG inequality, we have

- If A, B are increasing events, then

$$\mathbb{P}(AB) \geq \mathbb{P}(A)\mathbb{P}(B)$$

- If A, B are respectively increasing&decreasing events, then

$$\mathbb{P}(AB) \leq \mathbb{P}(A)\mathbb{P}(B)$$

<u>Question</u>: If A, B, C are all increasing, do we have $\mathbb{P}(AB|C) \geq \mathbb{P}(A|C)P(B|C)$

<u>Solution</u> Our answer is "NO". Here is an counterexample.

Consider a simple Boolean lattice space $\{0,1\}^2$ where each component is independent and takes the value 1 with probability p. Define:

- **Increasing event** C: At least one component is 1, i.e., $C = \{(0,1),(1,0),(1,1)\}$.
- **Increasing event** A: The first component is 1, i.e., $A = \{(1,0),(1,1)\}$.

- **Increasing event** B: The second component is 1, i.e., $B = \{(0,1),(1,1)\}$.

Calculations:

1. $P(AB|C)$: The event AB corresponds to $(1,1)$, with probability p^2. The conditional probability is:

$$P(AB|C) = \frac{p^2}{2p(1-p)+p^2} = \frac{p}{2-p}.$$

2. $P(A|C)$ **and** $P(B|C)$: By symmetry, both equal:

$$P(A|C) = \frac{p(1-p)+p^2}{2p(1-p)+p^2} = \frac{1}{2-p}.$$

   Thus, their product is:

$$P(A|C)P(B|C) = \left(\frac{1}{2-p}\right)^2.$$

Comparison:

$$\frac{p}{2-p} \geq \frac{1}{(2-p)^2} \quad \Rightarrow \quad p(2-p) \geq 1.$$

For $p = 0.5$, the left side becomes $0.5 \times 1.5 = 0.75 < 1$, which clearly fails. At this value:

- $P(AB|C) = \frac{0.5}{1.5} \approx 0.333,$
- $P(A|C)P(B|C) \approx 0.444,$

This violates the inequality.

Application Let $\mathscr{H}$ be a family of subsets of $[n]$ such that for all $A, B \in \mathscr{H}$, $A \cap B \neq \varnothing$ and $A \cup B \neq [n]$. Then $|H| \leq 2^{n-2}$.

Proof

Let $\mathscr{F}$ be the "upward order ideal" generated by $\mathscr{H} : \mathscr{F} = \{S : \exists T \in H, T \subseteq S\}$. Let G be the "downward order ideal" generated by $\mathscr{H} : \mathscr{G} = \{S : \exists T \in H, S \subseteq T\}$. Then $\mathscr{H} \subseteq \mathscr{F} \cap \mathscr{G}$.
$|\mathscr{F}| \leq 2^{n-1}$ because $\mathscr{F}$ satisfies the property that $\varnothing \subset A \cap B$ for all $A, B \in \mathscr{F}$, and therefore $\mathscr{F}$ cannot contain any set and its complement.
Likewise, $|\mathscr{G}| \leq 2^{n-1}$ because $\mathscr{G}$ satisfies the property that $A \cup B \neq [n]$ for all $A, B \in \mathscr{G}$, and therefore $\mathscr{G}$ cannot contain any set and its complement.
Interpreting this in terms of the bits being distributed uniformly i.i.d, we have that $\Pr(\mathscr{F}) \leq 1/2$ and $\Pr(\mathscr{G}) \leq 1/2$. Since $\mathscr{F}$ is an increasing event and $\mathscr{G}$ a decreasing event, $\Pr(\mathscr{F} \cap \mathscr{G}) \leq 1/4$.

A generalization of FKG inequality:

- (FKG Inequality II) Let $X_1, \ldots, X_n$ be independent random variables. Suppose f_1, f_2, f_3, f_4 are non-negative functions on $\mathbb{R}^n$ satisfying

$$f_1(x)f_2(y) \leq f_3(x \wedge y)f_4(x \vee y)$$

where x, y are n-dimensional vectors and $\wedge$ means taking coordinate-wise minimum and $\vee$ likewise. Then we have

$$\mathbb{E}f_1\mathbb{E}f_2 \leq \mathbb{E}f_3\mathbb{E}f_4$$

- (FKG Inequality III) Suppose $\mathbb{P}$ is a probability measure with density $p = \dfrac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mu}$ where μ is a product measure.

Assume $p(x)p(y) \leq p(x \wedge y)p(x \vee y)$. If f, g are increasing and $\mathbb{P} - L^2$, then we have

$$\mathbb{E}_{\mathbb{P}}(fg) \geq \mathbb{E}_{\mathbb{P}}f \cdot \mathbb{E}_{\mathbb{P}}g$$

- (FKG Inequality IV) Γ is a finite distributive lattice, and μ is a positive measure on Γ satisfying that for all $x, y \in \Gamma$,

$$\mu(x \wedge y)\mu(x \vee y) \geq \mu(x)\mu(y).$$

Then, for all increasing functions f, g on Γ, the inequality $\mathbb{E}(fg) \geq \mathbb{E}f \cdot \mathbb{E}g$ holds.

<u>Application</u> (Ising Model)

We have a finite graph $G = (V, E)$. We define a probability measure on

$$\forall \sigma \in \{-1, 1\}^V$$
$$\mathbb{P}(\sigma) = \frac{1}{Z}\exp\left(\beta \sum_{(u,v)\in E} \sigma_u\sigma_v\right)$$

where Z is a normalization constant, and $\beta \geq 0$ is a constant relative to temperature.

<u>Question</u>: Does FKG inequality holds for such probability measure?

<u>Solution</u>

We only need to check

$$\mathbb{P}(\sigma)\mathbb{P}(\tau) \leq \mathbb{P}(\sigma \wedge \tau)\mathbb{P}(\sigma \vee \tau)$$
$$\forall \sigma, \tau \in \{-1, 1\}^V$$

Key calculation:

For each edge (u, v), the term in the exponent of the density ratio is: $\sigma_u\sigma_v + \tau_u\tau_v - (\sigma \wedge \tau)_u(\sigma \wedge \tau)_v - (\sigma \vee \tau)_u(\sigma \vee \tau)_v$. When σ and τ differ at both u and v, this term equals -4, leading to an overall negative contribution to the exponent. For other cases, the term is 0.

The lattice condition holds, which implies FKG inequality also holds for true.

<u>Application</u> (Potts model)

The only difference Potts model from Ising model is $\sigma \in [n]^V$.

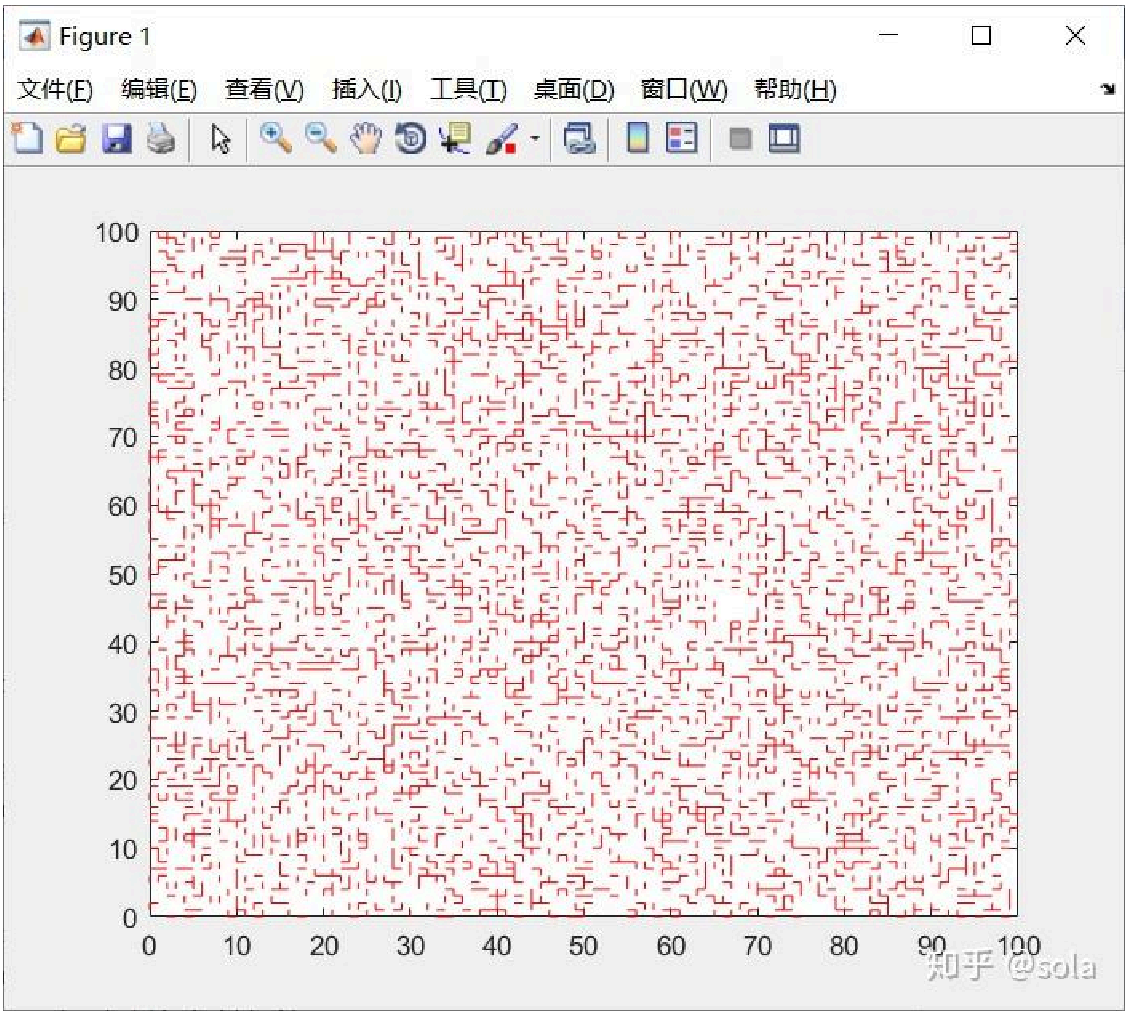
- (FKG Inequality for multivariable Gaussian) If X is multivariable Gaussian with $\Sigma = (\sigma_{ij})$ satisfying $\sigma_{ij} \geq 0$. Then

$$\mathbb{E}[f(X)g(X)] \geq \mathbb{E}f(X) \cdot \mathbb{E}g(X)$$

for increasing functions f, g.

Application (Percolation)

Our model is set on $\mathbb{Z}^d$, let each edge be open with probability p independently.

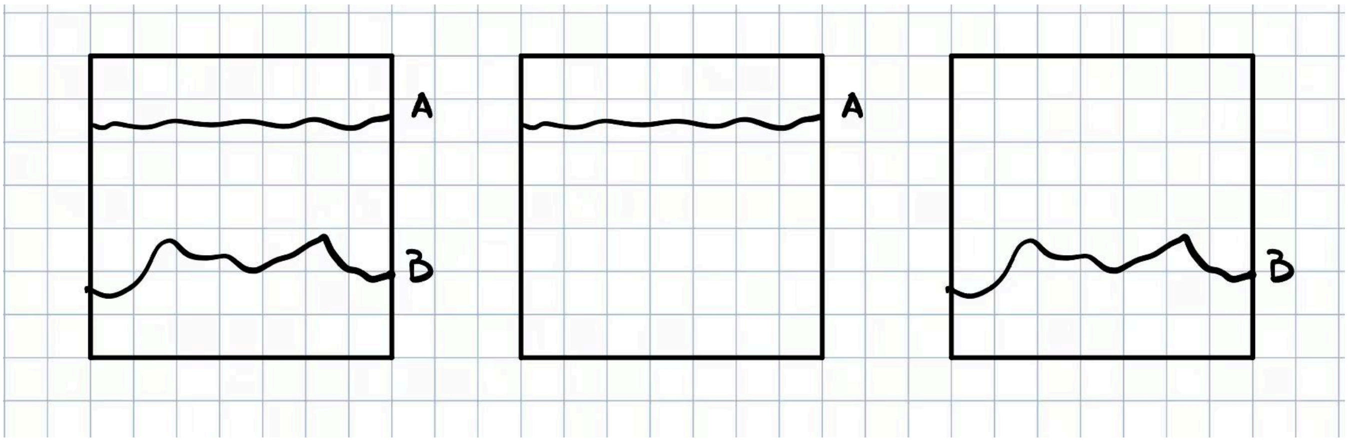


The figure shows an example on $\mathbb{Z}^2$.

And what we care about is the connected component generated by the open edges.

By FKG Inequality, we know that

means a rectangle can be connected from above or below.



Notice: We allow open edges in two parts to be overlapped.

Now we consider another events C: rectangle can be connected from above and simultaneously below with no edges overlapping.

Actually, we have

$$\mathbb{P}(C) \leq \mathbb{P}(A)\mathbb{P}(B)$$

(Intuition: local exploration)

To generalize, we want to construct an inequality that is directionally opposite to the FKG inequality.

Def $S \subset [n]$. An event A occurs on S in configuration w if $w \in A|_S$, where $A|_S = \{\omega : \forall \tilde{w} : \tilde{w} = w$ on S, we have $\tilde{w} \in A\}$.

Note: Knowing parts of the results on S, whether A occurs can be known.

Def Two events A_1, A_2 occur disjointly, denoted by $A_1 \circ A_2$, if there are two disjoint sets on which they occur:

$$A_1 \circ A_2 = \{\omega : \exists S_1, S_2 \subset [n], \mathrm{s.t.} S_1 \cap S_2 = \varnothing, \omega \in A_1|_{S_1} \cap A_2|_{S_2}\}$$

- (BKR Inequality) Consider $(\Omega, \mathscr{F}, \mu)$, where $\Omega = \Omega_1 \times \cdots \times \Omega_n, \mu = \mu_1 \times \cdots \times \mu_n$, then

$$\mu(A \circ B) \leq \mu(A)\mu(B)$$

First, let's see an application of BKR inequality in percolation.

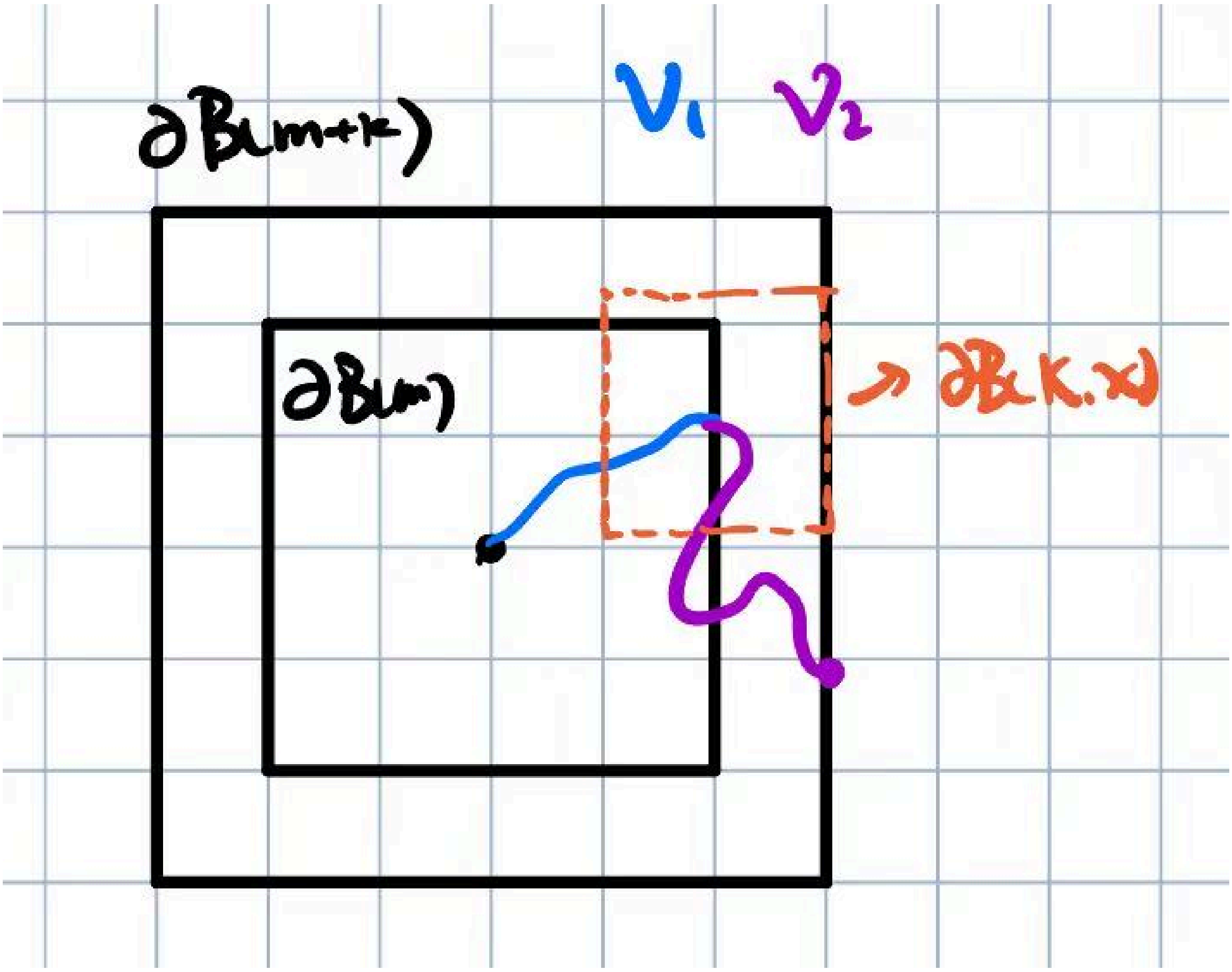
Let N be number of vertices in the open cluster containing origin O.

Suppose $\mathbb{E}N < \infty$, then $\mathbb{P}(O \leftrightarrow x) \leq e^{-c|x|}$ for some constant $c > 0$

A sketch of the proof

We have

$$O \leftrightarrow \partial B(m+k) \subseteq \bigcup_{x \in \partial B(m)} (O \leftrightarrow x) \circ (x \leftrightarrow \partial B(k, x))$$



Notice: We can require the path to be self-avoiding, and γ_1 and γ_2 not to be intersect. This assures "occur disjointly".

By BKR Inequality, we have

$$\begin{aligned}\mathbb{P}(O \leftrightarrow \partial B(m+k)) &\leq \sum_{x \in \partial B(m)} \mathbb{P}((O \leftrightarrow x) \circ (x \leftrightarrow \partial B(k,x)) \\ &\leq \sum_{x \in \partial B(m)} \mathbb{P}(O \leftrightarrow x)\mathbb{P}(x \leftrightarrow \partial B(k,x)) \\ &= \sum_{x \in \partial B(m)} \mathbb{P}(O \leftrightarrow x)\mathbb{P}(O \leftrightarrow \partial B(k)) \\ &= P(O \leftrightarrow \partial B(k))\mathbb{E}N_m\end{aligned}$$

where N_m is the number of nodes of $\partial B(n)$ to which the origin is connected.

Thus we can easily derive exponential decaying.